

languages

- formal
- informal

language is set of strings

example:

string start with 'a' & end with 'a'

• $L1 = \{aba, aa, ababa, abba, aabba, \dots\}$

Formal

predefined set of symbols & strings

→ concerned with rules (syntax) only (syntactic language)

→ not concerned with meaning.

Example:

language start with 'a' & ends with 'e'

formal language satisfied { ✓ ✓ }

a) Admhe

b) Amue

x c) nage

} meaningless but fulfills the rules.

→ meaning full but does not follow rules

Informal :-

→ concerned with rules & meaning (semantic language)

example:-

English language

✓ a) page → rule + meaning

x b) aegp

Basic Terminologies in Automata

> Letters characters or symbols

a, b, c, ...

0, 1, 2, 3, ...

> Alphabet

finite not empty set of letters

denoted by sigma Σ

$$\Sigma = \{a, b\}$$

alphabet contains only a & b letters

> String

concatenation of letters

if $\Sigma = \{a, b\}$

• ab, aaab, baba, babba,

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make a language for a machine string starts with 'b' and ends with 'b'

$$\Sigma = \{ a, b, c \}$$

$$L_1 = \{ bb, bab, baab, baccb, \dots \}$$

> word

string that is permissible in language

→ fulfills the condition.

only strings that fulfill given condition

> empty string

that has no letter, null string

$\wedge$ or λ

length of empty string will always be 0

> length of string

numbers of letter in a string

$|s|$

$$s = ababb \quad |s| = 5$$

$$\Sigma = \{ B, aB, bab, d \}$$

$$s = B aB bab B d$$

$$\text{Tokenizing} = (B) (aB) (bab) (B) (d)$$

$$|s| = 5$$

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> reverse of string

writing string in reverse order

Rev(s)

s = ababb

Rev(s) = bbaba

$\Sigma = \{a, aba, ca, d\}$

s = aabacada

tokenization s = (a)(aba)(ca)(d)

Rev(s) = d ca aba a

dcaabaa

> Power of alphabet

length of string = power

i) $\Sigma^2 = \{a, b\}^2$

$$\Sigma^2 = \{aa, ab, ba, bb\}$$

length is 2

ii) $\Sigma^3 = \{a, b, c\}^3$

$$\Sigma^3 = \{aaa, aba, abb, bca, cba, \dots\}$$

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> Power of String

$$S = aba \quad \Sigma$$

$$S^2 = (aba)^2 \\ = \underline{aba} \underline{aba}$$

$$S = a \boxed{b^2} a \\ = a \underline{bba}$$

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language whose reverse is same
as string

example:

$$S = \{abcba\} \quad \left. \begin{array}{l} \\ \text{rev}(S) = \{abcba\} \end{array} \right\} \text{ same}$$

Kleene Star

→ Kleene Closure

→ Kleene star Closure

denoted by Σ^*

“collection of all strings including null”

example :-

$$\text{if } \Sigma = \{a\}$$

$$\Sigma^* = ?$$

$$\Sigma^* = \{a\}^*$$

$$a^0 = \lambda$$

$$a^1 = a$$

$$a^2 = aa$$

$$a^3 = aaa$$

$$a^4 = aaaa$$

$$a^5 = aaaaa$$

⋮

$$a^n = n$$

$$\Sigma^* = \{\lambda, a, aa, aaa, aaaa, \dots\}$$

$$\text{ii) } \Sigma = \{a, b, c\}$$

$$\Sigma^* = ?$$

$$\Sigma^* = \{a, b, c\}^*$$

$$\{a, b, c\}^0 = \lambda$$

$$\{a, b, c\}^1 = \{a, b, c\}$$

$$\{a, b, c\}^2 = \{aa, ab, ba, ac, ca, bb, \dots\}$$

$$\{a, b, c\}^3 = \{aaa, aba, abc, cba, ccc, \dots\}$$

⋮

$$\{a, b, c\}^n = n$$

$$\Sigma^* = \{\lambda, a, b, c, ab, ba, abc, aab, abcb, abbca, \dots\}$$

$$\text{iii) } \Sigma = \{a, b\}$$

$$\Sigma^* = ?$$

$$\Sigma^* = \{\lambda, a, b, ab, bb, abb, bba, \dots\}$$

Kleene star in string

$$S = ab^*a$$

$$S = aa, aba, abba, abbbba, abbbbaa, \dots$$

Kleene Plus

→ kleene positive

→ positive closure

same as kleene star but
does not generate null
string.

always start with power 1

denoted by Σ^+

example

i) $\Sigma = \{a, b, c\}$

$\Sigma^+ = \{a, b, c, ab, cb, abc, \dots\}$

ii) kleene plus in string

$S = ab^+a$

$= aba, abba, abbbba, abbbbba, \dots$

Example

$\Sigma = \{0, 1\}$ find Σ^* & Σ^+

$\Sigma^* = \{\epsilon, 0, 1, 01, 00, 101, 100, 0001, \dots\}$

$\Sigma^+ = \{0, 1, 00, 11, 01, 100, 0001, \dots\}$

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 $\Sigma = \{ aaB, c \}$ find Σ^* & Σ^+
 $\Sigma^* = \{ \lambda, aaB, c, aaBc, aaBaaB, cc, \dots \}$
 $\Sigma^+ = \{ aaB, c, aaBc, cc, aaBaaB, aaBcc \}$

Defining languages

A language can be defined in following ways:-

- ∴ Descriptive
- ∴ Recursive
- ∴ Regular expression (RE)
- ∴ Finite Automata (FA)
- ∴ Transition Graphs (TG)

i) Descriptive Definition of language

describe condition imposed on its words
 → permissible strings

Example

- language L of string of even length defined over $\Sigma = \{ c \}$ $L = \{ cc, cccc, cccccc, \dots \}$

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2) language L of string that does not start with 'y'

defined over $\Sigma = \{x, y, z\}$

$L = \{x, z, xx, xy, zx, zy, zzx, zxy, zxyz\}$

3) language L of string of length 2,

defined over by $\Sigma = \{0, 1, 2\}$

$L = \{00, 01, 02, 11, 10, 12, 22, 20, 21\}$

4) language L of strings ending in 0

defined over $\Sigma = \{0, 1\}$

$L = \{0, 10, 00, 110, 100, 1010, 1110, \dots\}$

5) a language EQUAL, of strings

with number of a's equal to number of b's

$\Sigma = \{a, b\}$

$L = \{x, ab, aabb, abab, baab, baba, aaabbb, ababab, \dots\}$

6) a language EVEN-EVEN of strings with even number of a's & even number of b's
defined over $\Sigma = \{a, b\}$

$L = \{ \lambda, aa, bb, aabb, bbaa, abab, baba, bbbb, aaaa, abba, baab, aaaaabbbb, abababab, \dots \}$

7) a language $\{a^n b^n\}$ of string defined over $\Sigma = \{a, b\}$, as
 $\{a^n b^n : n = 1, 2, 3, 4, \dots\}$

$L = \{ab, aabb, aaabbb, aaaabbbb, \dots\}$

8) Write a language that contains atleast one 'a' defined over alphabet $\Sigma = \{a, s, d\}$

$L = \{a, as, sa, asd, sda, ads, adsa, aasd, \dots\}$

9) write a language that contains at exactly two 'cc' defined over alphabet $\Sigma = \{a, b, c\}$

$L = \{cc, accb, abcc, cca, ccb, bcc, \dots\}$

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10) a language start with "bb" &
end with 'c'

$$\Sigma = \{a, b, c, d\}$$

$$L = \{bbc, bbadc, bbac, bbabbd, \dots\}$$

Recursive Definition of language

three steps:

- 1) basic words are specified.
- 2) rules for constructing more words
- 3) No string except above two rules

example

i) defining language of positive even number
using recursive definition.

step 1: 2 is positive number

step 2: If x is positive even then so is, $x+2$

step 3: No string except those constructed above
are allowed to be in positive even

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2) Defining language of strings beginning ϵ ending in same letters define over $\Sigma = \{a, b\}$

step 1: a and b is in L

step 2: $(a)s(a)$ and $(b)s(b)$ are also in L , where s belongs to Σ^*

Step 3: No strings except those constructed in above, are allowed to be in L

3) Defining the language L , of strings containing 'aa' or 'bb', $\Sigma = \{a, b\}$

step 1: aa and bb are in L

step 2: $s(aa)s$ and $s(bb)s$ are also in L , where s belongs to Σ^*

Step 3: No strings except those constructed in above, are allowed to be in L

Regular Expression (RE)

language is represented in terms of strings.

examples:

1) alphabet $\Sigma = \{a\}$

language $L = \{a, aa, a aa, aaaa, \dots\}$

Regular Expression = RE = a^+

2) string start from 'a' & contain any 'b' letter defined over $\Sigma = \{a, b\}$

RE = $a b^*$

$L = \{a, ab, abb, abbb, abbbb, \dots\}$

3) string that start from 'a' and contains at least one 'b' letter $\Sigma = \{a, b\}$

RE = ab^+

$L = \{ab, abb, abbb, \dots\}$

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4) String that contains 'a' or 'b'
 $\Sigma = \{a, b\}$ $\downarrow$ union (+)

$$RE = a+b$$

5) string that contains any 'a' or any 'b'
 $\Sigma = \{a, b\}$
 $RE = (a+b)^*$

~~RE = (a+b)^*~~

6) string that contains at least one 'a' or at least 'b'
 $\Sigma = \{a, b\}$

$$RE = (a+b)^+$$

7) string that start with 'a' or end with 'a'
 $\Sigma = \{a, b\}$

$$RE = a(a+b)^*a$$

8) for the language accepting all strings starting with '1' and ending with '0'

$$\Sigma = \{0, 1\}$$

$$RE = 1(0+1)^*0$$

9) for language starting & ending with 'a' and having any combinations of 'b' in between $\Sigma = \{a, b\}$

$$RE = ab^*a$$

10) for the language $\Sigma = \{0, 1\}$

all strings do not contain substring 01

$$RE = (1^*0^*)$$

11) language containing the string in which every 0 is immediately followed by 11

$$RE = (011 + 1)^*$$

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12) for the language $\Sigma = \{a, b\}$
 words starting with 'a' & ending
 in 'b' OR starting with 'b' &
 ending in 'a'

~~RE = a(a+b)*b + b(a+b)*a~~

$$RE = a(a+b)^*b + b(a+b)^*a$$

Converting RE in language

$$L(a^* \cdot (a+b))$$

$$= L(a^*) L(a+b)$$

$$= (L(a^*) (L(a) \cup L(b)))$$

$$= \{ \epsilon, a, aa, aaa, aaaa, \dots \} \{a, b\}$$

$$= \{a, aa, aaa, \dots, b, ab, aab, \dots\}$$

✓

$$RE = (0+10^*)$$

$$L = \{0, 1, 10, 100, 1000, \dots\}$$